

Task 1

- Assumptions:
1. Customer Number uniquely identifies each customer. Which means, even if two customers have the same name, they still can be distinguished by customer number.
 2. There is no special relationship between specific employees and customers.
 3. The part name may not be unique, but part number is unique.
 4. All parts of the same type are associated with the same cage code. However, two different types of parts may share the same cage code.
 5. Each order can be uniquely identified by the combination of customerNumber, orderDate, time, employee and partNumber.
 6. One order can include multiple parts, and one part can appear in multiple orders.



- The **primary key** in 1NF should be (CustomerNumber, OrderDate, Time, PartNumber).
- CustomerNumber alone is not unique, since the same customer can place multiple orders.
 - OrderDate and Time alone are not unique, as many customers can place orders at the same time.
 - PartNumber determines PartName (since each part number corresponds to exactly one part name), but PartNumber alone is not unique in this context because a customer may order the same part multiple times or in different quantities.
 - However, the combination of CustomerNumber, OrderDate, Time, and PartNumber uniquely identifies one part ordered by a specific customer.

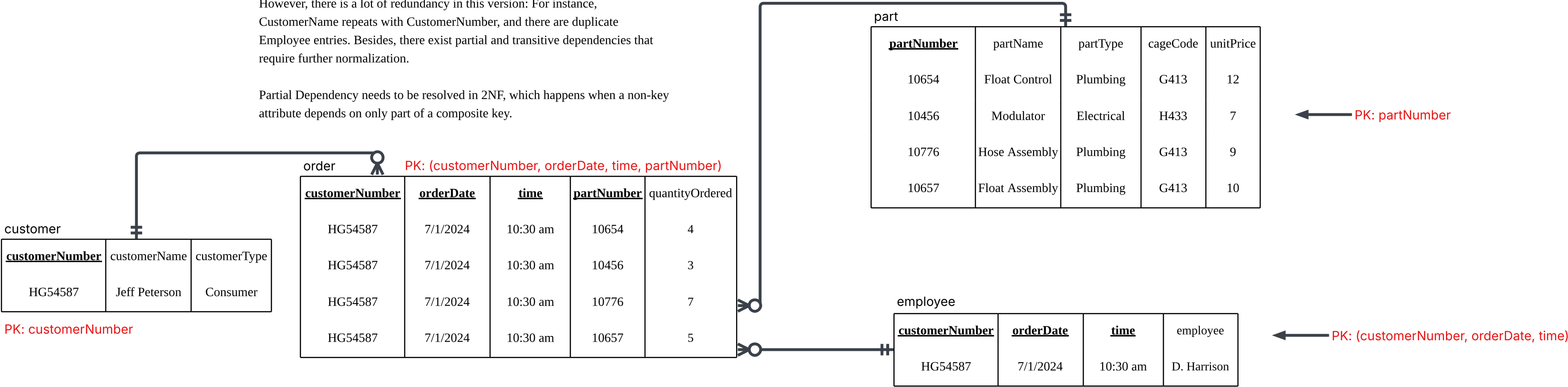


<u>customerNumber</u>	customerName	customerType	<u>orderDate</u>	<u>time</u>	employee	<u>partNumber</u>	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

This database is in **1NF** since each of its columns contains only atomic values, and there are no repeating groups of columns or rows.

However, there is a lot of redundancy in this version: For instance, CustomerName repeats with CustomerNumber, and there are duplicate Employee entries. Besides, there exist partial and transitive dependencies that require further normalization.

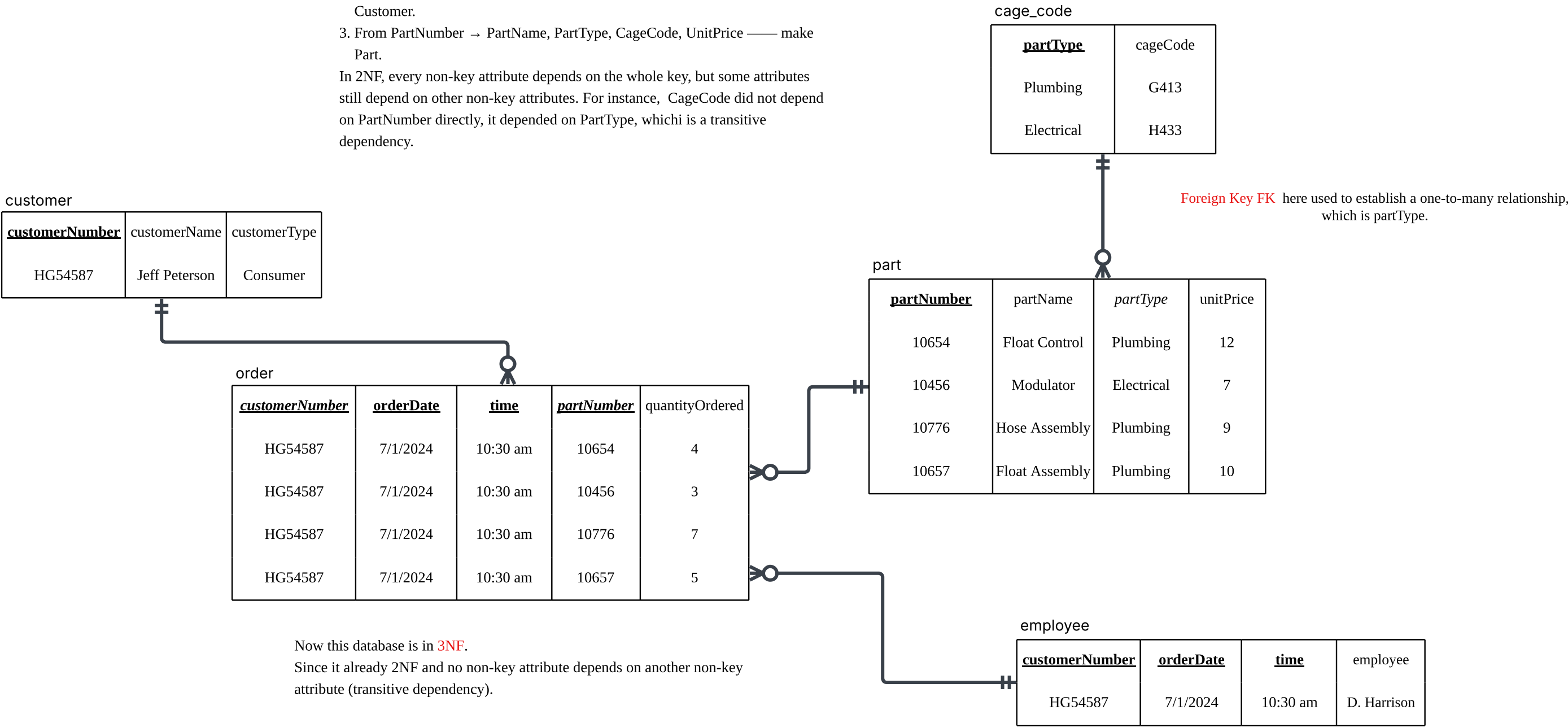
Partial Dependency needs to be resolved in 2NF, which happens when a non-key attribute depends on only part of a composite key.



This database is now in **2NF** since it is in 1NF and all the non-PK fields are dependent on All the PK fields.

- What I split:
1. From (CustomerNumber, OrderDate, Time, PartNumber) → Employee, QuantityOrdered — make Order .
 2. From CustomerNumber → CustomerName, CustomerType — make Customer.
 3. From PartNumber → PartName, PartType, CageCode, UnitPrice — make Part.

In 2NF, every non-key attribute depends on the whole key, but some attributes still depend on other non-key attributes. For instance, CageCode did not depend on PartNumber directly, it depended on PartType, which is a transitive dependency.



Now this database is in **3NF**. Since it already 2NF and no non-key attribute depends on another non-key attribute (transitive dependency).

Task 2

- Assumptions:
1. Each therapist has a unique staffNo and work at one or more branches.
 2. Each patient has a unique patNo.
 3. branchNo uniquely identifies a branch.
 4. Each appointment is for a specific date and time at a branch with a therapist.



staffNo	therapistName	patNo	patName	appointmentDateTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022 10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022 12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022 10:00	Q10
S1024	Heidi Pierce	P108	Andy Mckee	9/14/2022 14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022 16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022 18:00	B13

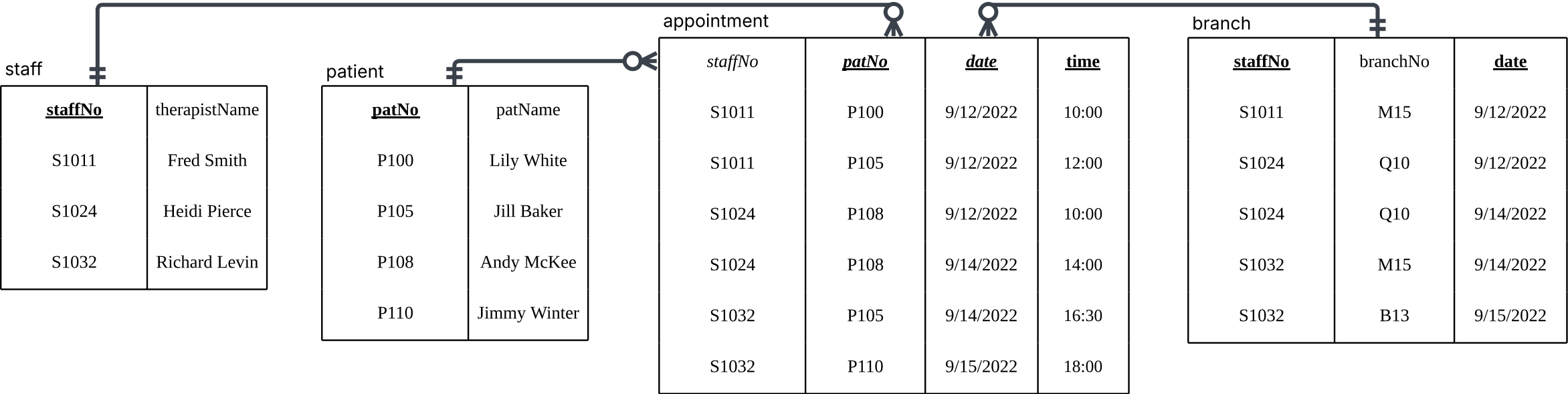
← This original table is not in 1NF yet, since in appointmentDateTime, multiple values in one field of a record.

<u>staffNo</u>	therapistName	<u>patNo</u>	patName	<u>date</u>	<u>time</u>	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy Mckee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

After splitting appointmentDateTime into date and time, this database is now in **1NF** since each of its columns contains only atomic values, and there are no repeating groups of columns or rows.

The **primary key** in 1NF should be (staffNo, patNo, date, time), which uniquely represents each appointment.

- However, redundancy exists:
1. therapistName repeats for the same staffNo. staffNo → therapistName
 2. patName repeats for the same patNo. patNo → patName
 3. branchNo is depends on staffNo and date since the therapist only see patients at one specific branch on any given day.



The table is currently in **3NF**. I have divided it into four parts: staff, patient, appointment, and branch. The primary key (**PK**) for staff is staffNo, for patient it is patNo, for appointment it is (patNo, date, time), and for branch it is (staffNo, date). Here, patNo and staffNo also serve as foreign keys (**FK**).

Task 3

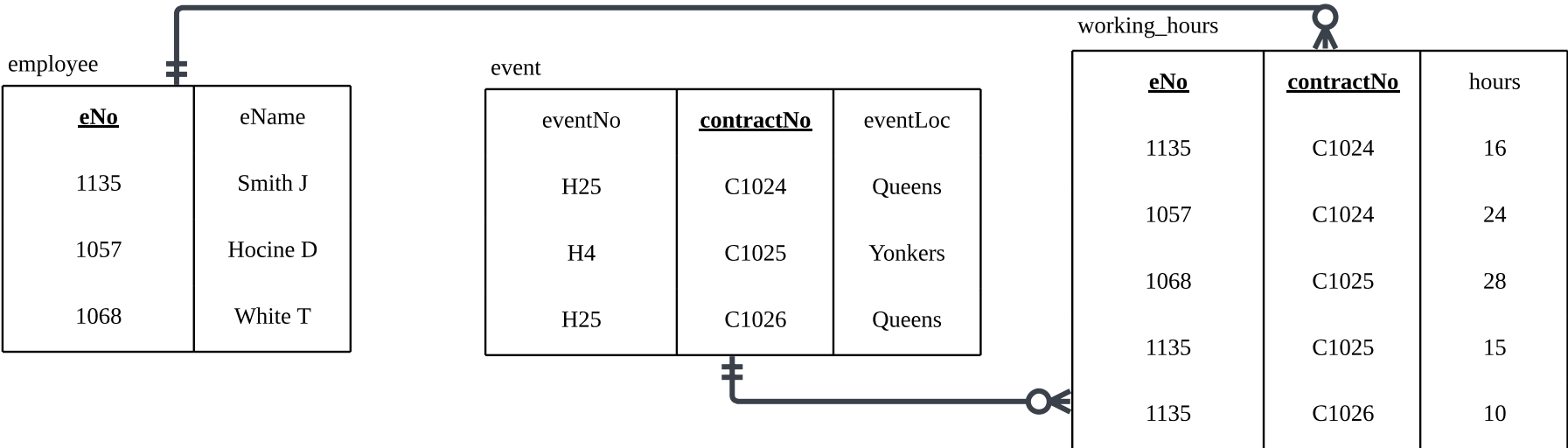
- Assumptions:
- 1. Each eNo uniquely identifies an employee, and each employee has exactly one eName.
 - 2. Each contractNo applies to exactly one eventNo.
 - 3. Each eventNo corresponds to exactly one eventLoc.
 - 4. hours represents the number of hours an employee worked on a specific contract.
 - 5. An employee may work on multiple contracts for the same event.

<u>eNo</u>	<u>contractNo</u>	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

This database is in **1NF** since each of its columns contains only atomic values, and there are no repeating groups of columns or rows.

Primary Key for this database is (eNo, contractNo), which uniquely identify each record.

However, partial dependency needs to be resolved in 2NF, which happens when a non-key attribute depends on only part of a composite key. For instance, eName depends only on eNo,. Similarly, eventNo and eventLoc depend only on contractNo.

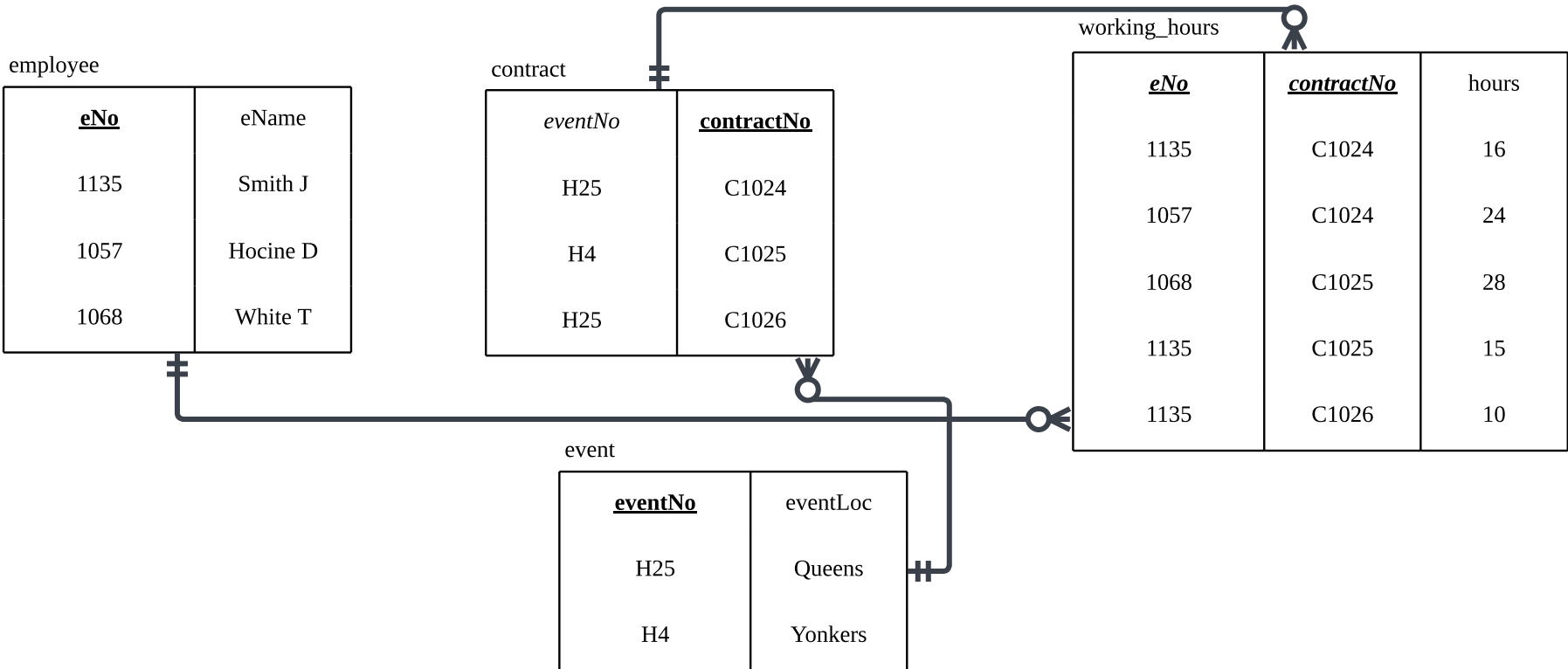


This database is currently in **2NF** because all partial dependencies have been resolved. Each non-key attribute now fully depends on the whole primary key of its relation.

However, the database is not yet in 3NF, because there is still a transitive dependency. Specifically, eventLoc does not directly depend on the primary key of the Contract relation (contractNo), but rather indirectly through eventNo.

To achieve 3NF, we need to further decompose the tables to eliminate this transitive dependency.

After this split, all non-key attributes will depend only on their relation’s primary key, satisfying 3NF.



In the Contract table, eventNo now serves as a **foreign key FK** referencing the Event table.